National
Qualifications
2026

X847/76/11

Mathematics
Paper 1 (Non-calculator)



THURSDAY, 7 MAY
9:00 AM – 10:15 AM

Total marks — 55

Attempt ALL questions.

You must NOT use a calculator.

To earn full marks you must show your working in your answers.

State the units for your answer where appropriate.

You will not earn marks for answers obtained by readings from scale drawings.

Write your answers clearly in the spaces provided in the answer booklet. The size of the space provided for an answer is not an indication of how much to write. You do not need to use all the space.

Additional space for answers is provided at the end of the answer booklet. If you use this space you must clearly identify the question number you are attempting.

Use **blue** or **black** ink.

You must leave your answer booklet on your desk; if you do not, you could lose all the marks for this paper.

Qualifications Scotland
Teisteanasan Alba

X8477611

FORMULAE LIST

Circle

The equation $x^2 + y^2 + 2gx + 2fy + c = 0$ represents a circle centre $(-g, -f)$ and radius $\sqrt{g^2 + f^2 - c}$.

The equation $(x-a)^2 + (y-b)^2 = r^2$ represents a circle centre (a, b) and radius r.

Scalar product $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos\theta$, where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$

or $\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3$ where $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$.

Trigonometric formulae

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$= 2\cos^2 A - 1$$

$$= 1 - 2\sin^2 A$$

Table of standard derivatives

$f(x)$	$f'(x)$
$\sin ax$	$a \cos ax$
$\cos ax$	$-a \sin ax$

Table of standard integrals

$f(x)$	$\int f(x)dx$
$\sin ax$	$-\frac{1}{a}\cos ax + c$
$\cos ax$	$\frac{1}{a}\sin ax + c$

MARKS

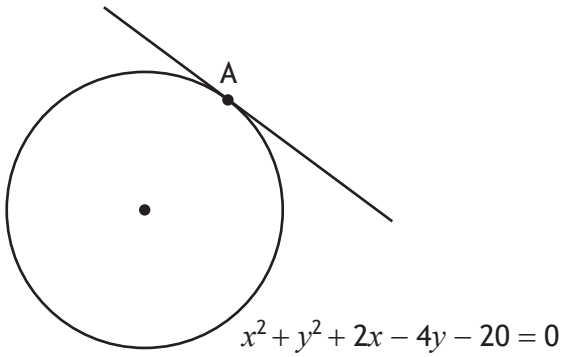
Total marks — 55

Attempt ALL questions

1. Express $2x^2 + 20x + 3$ in the form $p(x+q)^2 + r$. **3**

2. Find $\int\left(15x^{\frac{2}{3}} + \frac{7}{x^2}\right)dx,\ x > 0.$ **4**

3. A circle has equation $x^2 + y^2 + 2x - 4y - 20 = 0$.
Point A(2,6) lies on the circle.

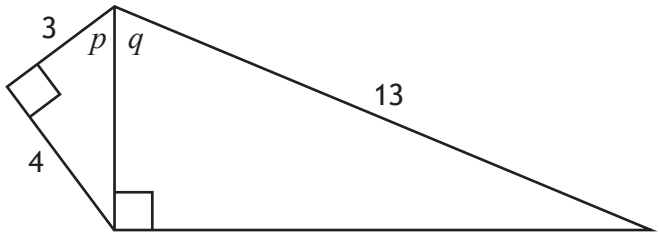


Find the equation of the tangent to the circle at A. **4**

[Turn over

MARKS

4. The diagram shows two right-angled triangles with angles p and q as marked.



(a) Determine the exact value of $\sin q$. **1**

(b) Find the exact value of:

(i) $\sin(p+q)$ **4**

(ii) $\cos(p+q)$. **2**

(c) Hence find the exact value of $\tan(p+q)$. **1**

5. Functions f and g are defined on $\mathbb{R}$, the set of real numbers, by $f(x)=2x^2+5$ and $g(x)=x+3$.

(a) Find an expression for:

(i) $f(g(x))$ and **2**

(ii) $g(f(x))$. **1**

(b) State the range of $g(f(x))$. **1**

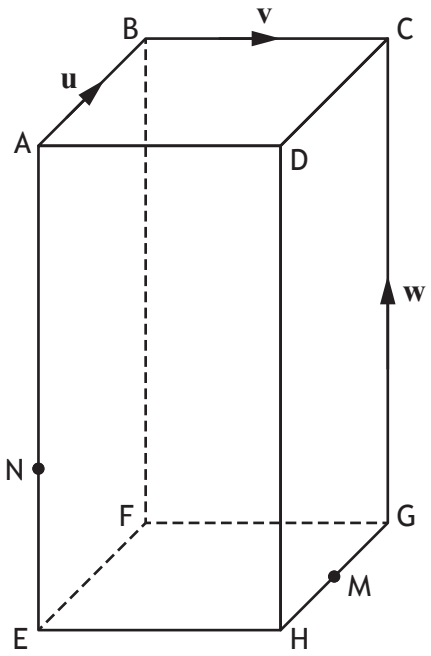
MARKS

6. ABCD,EFGH is a cuboid.

$\overrightarrow{AB} = \mathbf{u}$, $\overrightarrow{BC} = \mathbf{v}$ and $\overrightarrow{GC} = \mathbf{w}$.

The point M is the mid-point of HG.

The point N divides EA with the ratio 1:2.



Express the vector $\overrightarrow{MN}$ in terms of $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$. **2**

7. Determine the gradient of the tangent to the curve with equation $y = 2x + 4\sqrt{x}$, $x > 0$, at the point where $x = 9$. **4**

8. Solve $\log_2 x + \log_2 (x+2) = 3$, where $x > 0$. **4**

[Turn over

MARKS

9. (a) Show that the points D(−1,6,1), E(1,2,7) and F(2,0,10) are collinear. **3**

Point G is such that F divides DG in the ratio 3:4.

(b) Find the coordinates of G. **2**

10. Find $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} 6\sin\left(3x-\frac{\pi}{2}\right)dx$. **4**

11. (a) (i) Show that $(x+2)$ is a factor of $x^3+7x^2+18x+16$. **2**

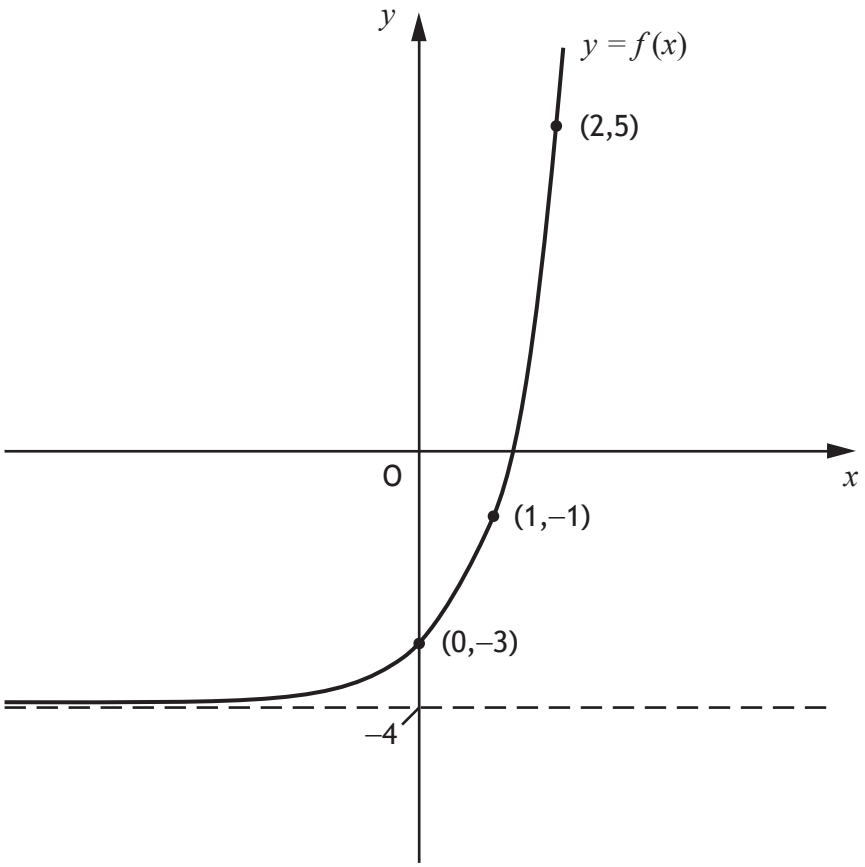
(ii) Explain why $(x+2)$ is the only linear factor of $x^3+7x^2+18x+16$.
Give a reason for your answer. **2**

(b) Find the coordinates of the point(s) where the curves with equations $y=2x^3+20x^2+27x+9$ and $y=6x^2-9x-23$ intersect. **4**

MARKS

12. An exponential function, f, is defined for $x \in \mathbb{R}$.

The diagram shows the graph of $y = f(x)$.



The inverse function, $f^{-1}(x)$, exists.

(a) On the diagram in your answer booklet, sketch the graph of the inverse function. **2**

The inverse function is of the form $f^{-1}(x) = \log_a(x+b)$.

(b) (i) Determine the values of a and b. **2**

(ii) State the domain of the inverse function, $f^{-1}(x)$. **1**

[END OF QUESTION PAPER]

[BLANK PAGE]

DO NOT WRITE ON THIS PAGE